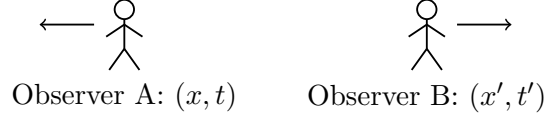


Lorentz transformation



Consider two inertial observers with parallel spatial axes. B moves at velocity $+v$ relative to A; A moves at $-v$ relative to B. The arrows indicate these reciprocal motions. Let their origins coincide at $t = t' = 0$, and use identical units and synchronised clocks within each frame.

1. Linearity

Imagine a rope divided into adjacent lengths l_1 and l_2 . Its measured length must be additive:

$$T(l_1 + l_2) = T(l_1) + T(l_2).$$

Together with continuity, this implies a linear length map. For the full spacetime transformation, homogeneity of space and time gives the corresponding additivity of coordinate intervals. Continuity and the common origin then give a linear transformation.

The same event, such as a flash, has coordinates (x, t) and (x', t') . Write

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = M \begin{pmatrix} x \\ t \end{pmatrix}, \quad M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad d = \det M = AD - BC.$$

Here A, B, C, D are coefficients depending on v , distinct from the observer labels. The inverse is

$$\begin{pmatrix} x \\ t \end{pmatrix} = \frac{1}{d} \begin{pmatrix} D & -B \\ -C & A \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix}.$$

2. Follow each observer's origin

At B's origin, $x' = 0$ and $x = vt$. Hence

$$0 = Ax + Bt = (Av + B)t \implies B = -Av.$$

At A's origin, $x = 0$ and $x' = -vt'$. The inverse transformation gives

$$0 = Dx' - Bt' = (-Dv - B)t' \implies B = -Dv.$$

Thus $A = D$ for $v \neq 0$. The case $v = 0$ is obtained continuously, with $M(0) = I$.

3. Clock symmetry and the determinant

For two ticks of A's clock, $\Delta x = 0$, so

$$\Delta t' = D \Delta t.$$

For two ticks of B's clock, $\Delta x' = 0$, so the inverse gives

$$\Delta t = \frac{A}{d} \Delta t'.$$

These refer to two different pairs of events. By the principle of relativity and spatial isotropy, exchanging the observers gives the same clock-rate factor. Therefore $D = A/d$. Since $A = D$, we obtain $d = 1$.

It follows that

$$AD - BC = A^2 + AvC = 1 \implies C = \frac{1 - A^2}{Av}.$$

Our transformation is now

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} A & -Av \\ \frac{1 - A^2}{Av} & A \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}.$$

4. Use the invariance of the speed of light

Let a light ray travel in the positive direction from the common origin. Both observers measure its speed as c , so $x = ct$ and $x' = ct'$. Substitution gives

$$ct' = A(c - v)t, \quad t' = (Cc + A)t.$$

Eliminating t' ,

$$c(Cc + A) = A(c - v) \implies Cc^2 = -Av.$$

Using the expression for C ,

$$\frac{1 - A^2}{Av} c^2 = -Av \implies (1 - A^2)c^2 = -A^2 v^2 \implies A^2(c^2 - v^2) = c^2.$$

Choose the positive root, continuously connected to $A(0) = 1$:

$$A = \frac{1}{\sqrt{1 - v^2/c^2}} \equiv \gamma, \quad C = -\frac{\gamma v}{c^2}, \quad |v| < c.$$

5. The Lorentz transformation and its inverse

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -v \\ -v/c^2 & 1 \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}, \quad \begin{pmatrix} x \\ t \end{pmatrix} = \gamma \begin{pmatrix} 1 & v \\ v/c^2 & 1 \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix}.$$

Equivalently,

$$\begin{aligned} x' &= \gamma(x - vt), & t' &= \gamma\left(t - \frac{vx}{c^2}\right), \\ x &= \gamma(x' + vt'), & t &= \gamma\left(t' + \frac{vx'}{c^2}\right). \end{aligned}$$

Reference: *Upgrade Your Physics*, revision April 2017, §2.4.1, pp. 31–33. The sequence follows the handwritten notes, with brief clarifications and corrected signs.